LORA ROOM SERVER

LoRaMesh

OFFLINE • STRUCTURED • CHAT ROOMS

Structured. Persistent. Off-Grid.

Beyond Standard LoRa

Standard Meshtastic networks act like walkie-talkies. Our "Room Server" acts as an intelligent backend, bringing the power of **modern chat** applications (like Discord or Slack channels) to off-grid radio networks. It provides persistent memory, structured rooms, and robust data storage entirely **without the internet**.

Project Details & Tech Stack

Developed by: Arhan UNAY & Adam TAWFIK

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Polytech Grenoble - INFO4 (2025-2026)

Tech Stack:

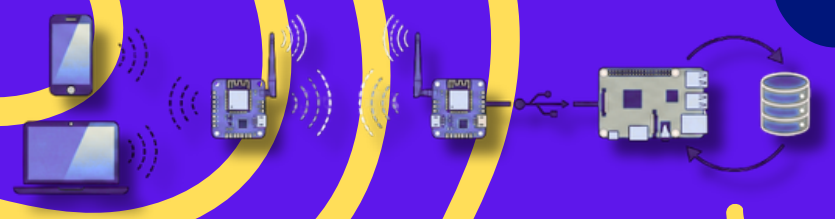
Scan for Source Code

Scan for Documents & Report

GRENOBLE INP UGA

POLYTECH GRENOBLE

How It Works

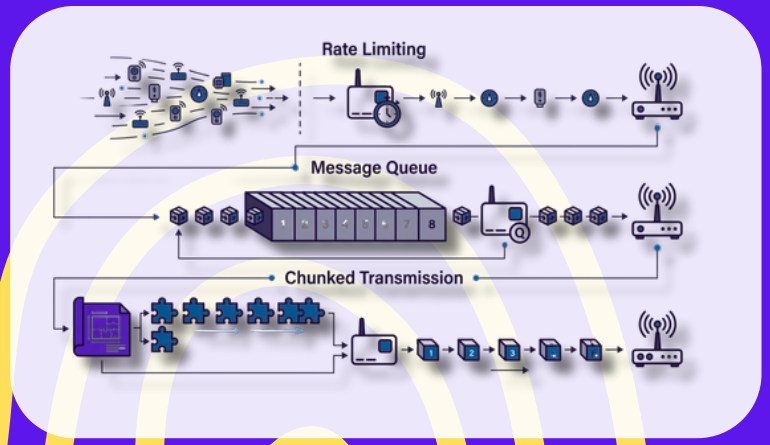


A **Raspberry Pi** acts as the server's brain, connected to a **Wio-E5 LoRa** module. Users send commands via their smartphones (connected via Bluetooth or via cable to their own Client LoRa nodes). The server interprets these commands, queries its local **SQLite** database, and transmits the response back through the radio waves.

Send commands via Direct Message (DM) to the server node:

Create a room: /room create <name>
Delete a room: /room delete <name>
List all rooms: /room list
Post a message: /room post <name> <msg>
Read history: /room read <name>
Broadcast message: /room announce <msg>
Show help: /room help or /room ?
Info about a room: /room info <name>
Broadcast message: /room announce <msg>

Engineering for LoRa

Radio waves are slow and heavily restricted. We **optimized** the system with:

- **Anti-Spam (Rate Limiting):** A 10-second cooldown per user to protect network Duty Cycle.
- **FIFO Queueing:** Prevents hardware from crashing when receiving multiple requests simultaneously.
- **Chunking:** Responses are cut into 32-character blocks to guarantee delivery over long distances.